



AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

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7.Configure SFTP toLinux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php, MySQL or other Linux-based applications.

SFTP- Secure File Transfer Protocol

SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

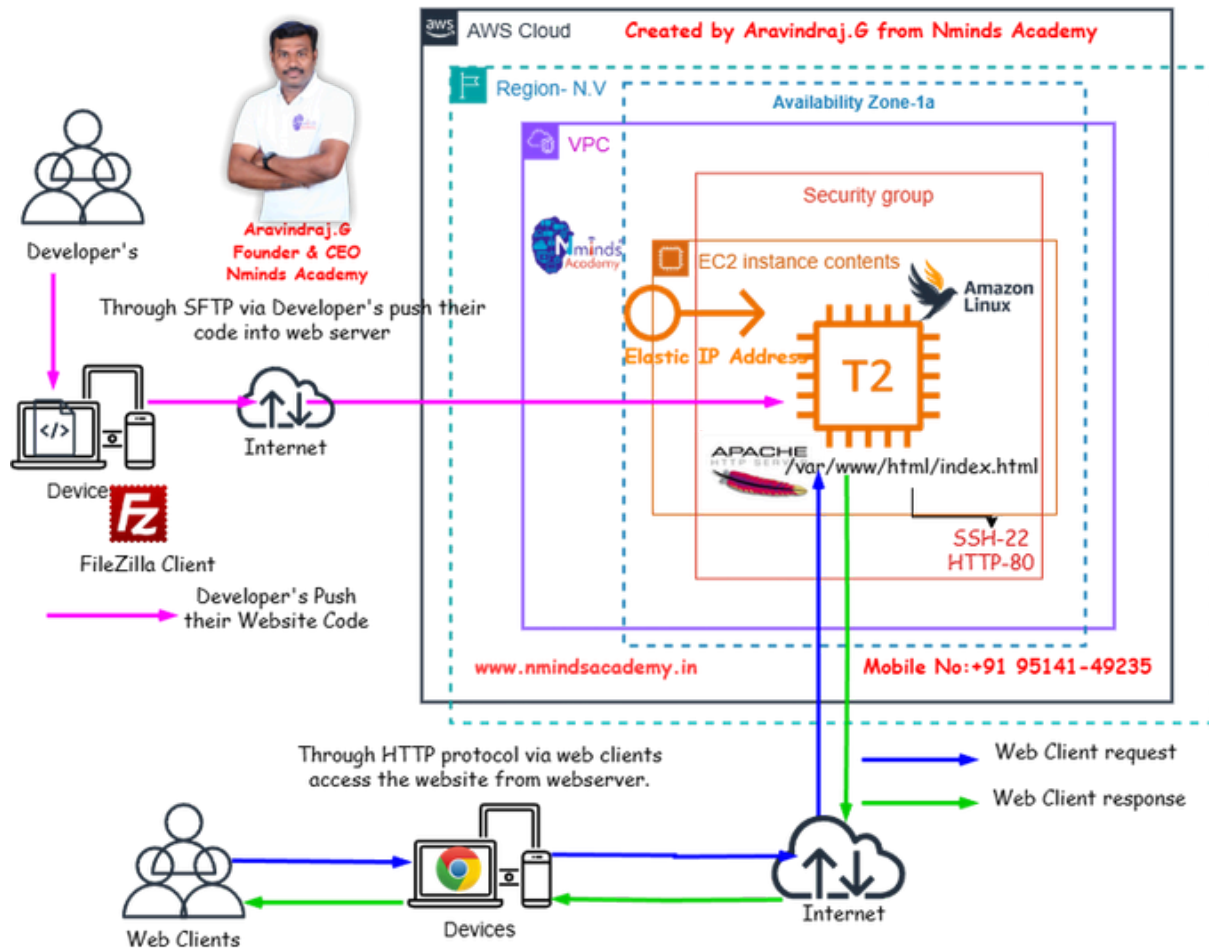
Here are the key uses and advantages of SFTP:

1. **File Transfer:** SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. **File Management:** Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. **Secure Authentication:** SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. **Encryption:** The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. **Firewall-Friendly:** SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.



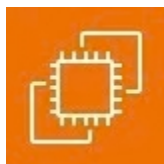
Topology

How to Configure Developer's push their website code by using SFTP Protocol in Linux Server with AWS EC2 with Elastic IP Address



Used Services in AWS
Services in Windows Server

Used



Amazon EC2



Apache Webserver



Amazon Elastic IP Address



Execution Tasks:

```

ec2-user@ip-172-31-16-241:~ + ~
w] 50L, 12918 written
t@ip-172-31-16-241 html]# client_loop: send disconnect: Connection reset

sers\shabh\Downloads>ssh -i Shabahri.pen ec2-user@44.210.222.117
ing: Identity file Shabahri.pen not accessible: No such file or directory.
authenticity of host '44.210.222.117 (44.210.222.117)' can't be established.
519 key fingerprint is SHA256:SUKMwbNsfnKwE5ejPEpDQxVmnqdHtg316T3QoZs1o/Y.
host key is known by the following other names/addresses:
C:\Users\shabh/.ssh/known_hosts:1: 54.226.63.79
you sure you want to continue connecting (yes/no/[fingerprint])? yes
ing: Permanently added '44.210.222.117' (ED25519) to the list of known hosts.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pen ec2-user@44.210.222.117
ing: Identity file Shabhari.pen not accessible: No such file or directory.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pem ec2-user@44.210.222.117
#_
\_ #####_ Amazon Linux 2023
\_#####\
\_###|
\_#/ --- https://aws.amazon.com/linux/amazon-linux-2023
V~' ' ->
login: Thu Mar 12 18:05:49 2026 from 171.79.58.49
-user@ip-172-31-16-241 ~]$ |

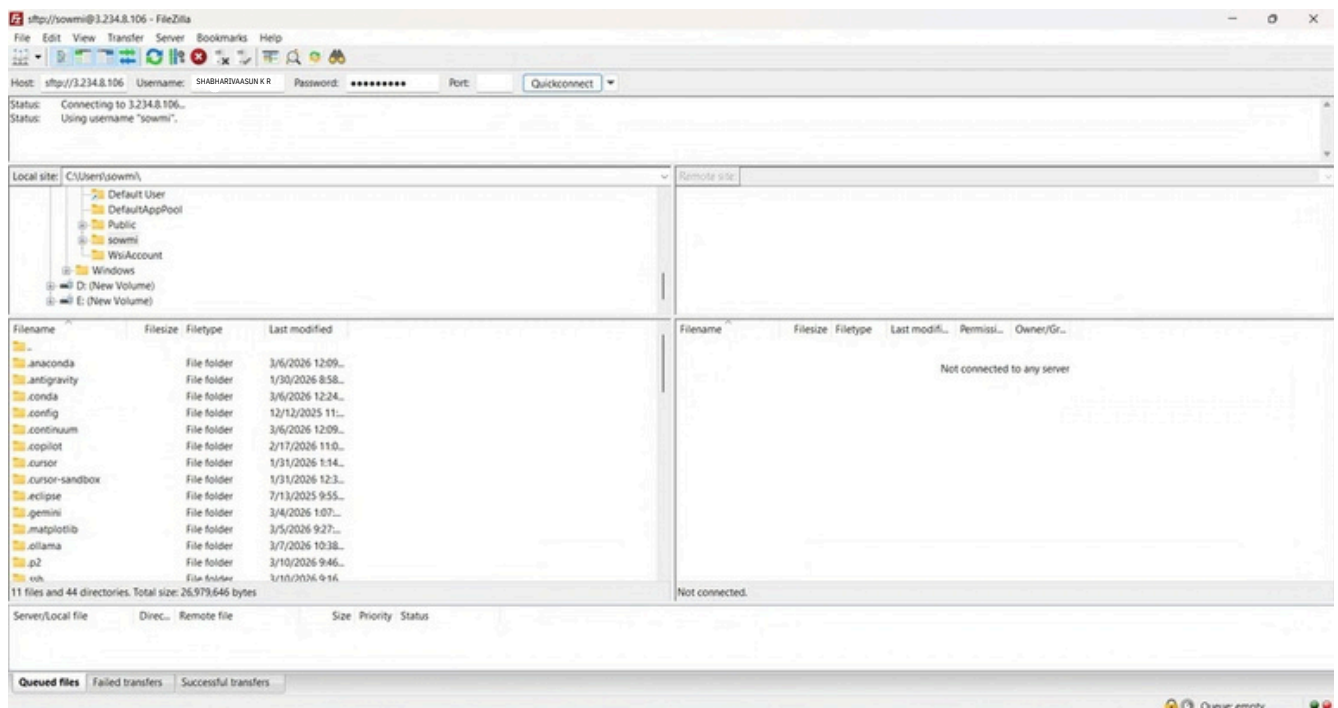
```

```
root@ip-172-31-16-241:~  
daemon:x:2:2:daemon:/sbin:/sbin/nologin  
adm:x:3:4:adm:/var/adm:/sbin/nologin  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
sync:x:5:0:sync:/sbin:/bin/sync  
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown  
halt:x:7:0:halt:/sbin:/sbin/halt  
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin  
operator:x:11:0:operator:/root:/sbin/nologin  
games:x:12:100:games:/usr/games:/sbin/nologin  
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin  
nobody:x:65534:65534:Kernel Overflow User:/sbin/nologin  
dbus:x:81:81:System message bus:/sbin/nologin  
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin  
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin  
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin  
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin  
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin  
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin  
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin  
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin  
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin  
ec2-instance-connect:x:993:993:./home/ec2-instance-connect:/sbin/nologin  
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin  
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin  
tcpdump:x:72:72:./:/sbin/nologin  
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash  
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin  
dev1deeps:x:1001:1001:./home/dev1deeps:/bin/bash  
dev2deeps:x:1002:1002:./home/dev2deeps:/bin/bash  
root@ip-172-31-16-241 ~]#
```



Execution Tasks:

```
root@ip-172-31-72-105:~  
# Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
#LoginGraceTime 2m  
#PermitRootLogin prohibit-password  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
PubkeyAuthentication no  
  
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2  
# but this is overridden so installations will only check .ssh/authorized_keys  
AuthorizedKeysFile .ssh/authorized_keys  
  
#AuthorizedPrincipalsFile none  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# Explicitly disable PasswordAuthentication. By presetting it, we  
# avoid the cloud-init set_passwords module modifying sshd_config and  
# restarting sshd in the default instance launch configuration.  
PasswordAuthentication yes  
PermitEmptyPasswords no  
  
# Change to no to disable s/key passwords  
#KbdInteractiveAuthentication yes  
  
# Kerberos options  
#KerberosAuthentication no  
-- INSERT --
```



Execution Tasks:

